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Hybrid deep learning framework for heart disease prediction using ECG signal images

Sivabalaselvamani Dhandapani[1]✉, Hemalatha Somasundaram[2] & Tamilarasi Angamuthu[2]

With cardiovascular diseases accounting for all other causes of mortality worldwide, an increasing proportion of individuals are being treated for them. To identify the cardiac issue, medical practitioners have to examine electrocardiogram (ECG) data. Moreover, diagnosis is a process that healthcare professionals find simple vulnerable to mistakes. Modern deep learning systems have tackled the difficult choreography of independently identifying heart disease using ECGs. The present study is in line with the increasing frequency of cardiovascular disease, a major cause of death and sickness globally. From earlier and more precise detection of cardiac issues, improved patient outcomes and lower healthcare costs are conceivable. Although electrocardiogram (ECG) impulses are a diagnostic tool of great significance, their interpretation usually depends on professional analysis, which is prone to human variability and error. With an aim to lower computational complexity, this study proposes a hybrid deep learning framework for heart disease prediction utilizing artificial neural network models. This work aims to develop a deep learning approach using artificial neural networks (DLA-ANNs), automatically diagnoses heart disorders. The data have come from an ECG heartbeat classification kaggle dataset. The experimental results reveal that the ANN-based design provides superior accuracy than the state-of- the-art approaches. The outcomes reveal that the recommended strategy performs effectively, so it might be implemented in a medical setting. The proposed method surpasses other previously in use methods in several important respects: accuracy (93.6%), sensitivity (97.4%), adaptability (98.2%), performance (97.9%), and scalability (96.8%).

Keywords Electrocardiogram (ECG), Heart disease prediction, Artificial neural network (ANN), Deep learning framework, Cardiovascular health, Medical diagnostics

The complexity and unexpectedness of ECG data makes it difficult to create a hybrid deep-learning system that can predict heart disease using images of ECG signals[1]. Sophistidated deep-learning models that can capture and understand the complex content of electrocardiogram (ECG) data are required for reliable diagnosis of heart diseases[2]. The creation of a hybrid framework that effectively merges RNN for sequence modeling with firm nervous network (CNNS) for picture interpretation, does not lack its technical challenges, but[3]. Due to the sequential nature of data represented by the electrical activity of the heart throughout the time, special design is necessary to capture temporary relationships and long distance dependence in the electrocardiogram (ECG) signals[4]. The model is generally more interrupted by the fact that noise, artifacts and differential ECG signal images in the electrode location can vary greatly in quality and stability[5]. There may be a lack of ECG dataset labeled for training and testing, which can increase the possibility of prejudice and overfiting during model construction[6]. Clinical adoption and belief among healthcare physicians should ensure that the predictions of the hybrid deep learning models can be explained and understood[7]. In order to remove these obstacles and guarantee the dependence and therapeutic relevance of the prediction of heart disease, it is necessary to collaborate with domain experts and to carefully employ algorithm design, data pre-processing methods, verification procedures and cooperation[8]. Efforts should be made to standardize the methods of acquisition, promote data sharing and improve data quality to increase the accuracy and reliability of intensive learning models for ECG-based heart disease prediction[9]. The prophecy of cardiac disease using ECG signal images often uses hybrid deep learning framework that combines CNN for image analysis with RNN for sequence modeling[10]. While RNN sequential ECG data is great in detecting temporary connections, CNNS ECG removes spatial characteristics from signs[11], which can reveal patterns of various cardiac problems[12]. The prediction model of heart disease is obtained using better accuracy and flexibility hybrid architecture that allows for intensive examination of spatial and cosmic

[1]School of Information Science, Presidency University, Bengaluru, India. [2]Department of Computer Applications, Kongu Engineering College, Perundurai, Erode, India. ✉email: sivabalaselvamani@presidencyuniversity.in

characteristics of ECG signals[13]. It is a common practice that transfers to learning to learn, data growth, and model performance to various patient population[14].

Hybrid deep learning framework electrocardiogram (ECG) promises to predict heart disease with signal images, but they face many obstacles[15]. Potential bias and overfiting problems can arise when deep learning models are trained on small scale anotate dataset, which is a major obstacle[16]. When confidence and openness are important in medical settings, the interpretation and interpretation of these complex models is still an issue[17]. Variations in noise, artifacts, and acquisition techniques may vary to ECG signal images in quality and stability, making it difficult to normalize the model and apply them to the real world[18]. In addition, low-recourse healthcare settings or mobile devices may not be able to apply deep teaching models due to their computational complexity and resource requirements. To overcome these obstacles, researchers, physicians, and data scientists should work together to make deep learning models that are both accurate and easy to understand, to standardize and improve data quality, and to ensure that ECG signal images are used to predict heart disease, they are adopted based on artificial intelligence methods. The inability of the current approach, error-intelligence, and dependence on expert expertise, such as manual ECG interpretation, obstructing timely treatment. Traditional methods require adequate facility engineering to adequately interpret ECG data due to their complexity. An encouraging development in the region is the rise of an artificial nervous network (ANNS), which can automatically learn patterns in ECG signal images without human participation. Better patient results in the fight against heart disease can be achieved by making DLA-Es, which aims to use these skills early, reduce clinical mistakes, and eventually, to empower personal healthcare by offering personal risk assessment.

The main objective of this research is to create a hybrid deep learning framework called DLA-Es that can predict the efficiency and accurately of the occurrence of cardiac disease using electrocardiogram (ECG) paintings. The purpose of the framework is to automate clinical procedures so that doctors do not need as much and patients can get care that they need fast. Among the achievements of the study is the DLA-Ann framework, which provides a scalable and reliable clinical solution by combining the sophisticated ECG image and fully connected nerve networks for the exact ECG image processing. At its top, a comprehensive evaluation shows the accuracy, sensitivity and uniqueness of the framework compared to the current functioning.

The primary contributions of this study are outlined as follows:

- Designing the deep learning method using artificial neural networks (DLA-ANNs) that integrates convolutional and fully connected layers for precise analysis of ECG signal images, advancing the state-of-the-art in heart disease prediction.
- Introducing the efficient framework capable of automating the interpretation of ECG signals, reducing reliance on manual expertise while ensuring scalability for large datasets.
- Conducting a detailed performance analysis of the proposed framework against existing methods, demonstrating superior accuracy, sensitivity, and specificity for heart disease prediction.
- A major limitation in these studies is the loss of critical temporal information when converting 1D ECG signals into 2D images, which can reduce the models ability to accurately capture time-dependent cardiac features essential for precise heart disease prediction.

The rest of the paper is structured as follows: "Literature survey" section reviews the research on Hybrid Deep Learning Frameworks for Cardiovascular Diseases. Deep learning algorithms that use artificial neural networks (DLA-ANNs) are mathematically discussed in "Deep learning algorithm based on artificial neural networks" section. "Results and discussion" section discusses experiment's results, analysis, and comparisons to previous methods. "Conclusion" section showcases the ultimate conclusions.

Literature survey

Numerous cutting-edge approaches have been developed in medical diagnostics in response to the growing need for precise and effective illness prediction and classification methods. DLA-ANNs, based on artificial neural networks, have been quite effective in accurately classifying diseases, especially heart disorders.

Fatema et al.[19] provide a deep learning-based approach (DL-A) for accurate cardiac disease classification using paper ECG images. Along with the five separate models, a hybrid model dubbed InRes-106 is introduced. It combines InceptionV3 with ResNet50. Achieving the best testing accuracy of 98.34%, the hybrid model outperformed separate models, according to the comprehensive experimentation. A hybrid model combining continuous wavelet transformation (CWT) and convolutional neural networks (CNN) is presented by Mohammad et al.[20] as a means of predicting heart disease using electrocardiogram (ECG) data. It achieves a remarkable 97.2% accuracy while handling dataset imbalances and extracting features using CWT, all with the help of RUSBoost.

Kaspal et al.[21] used Dropout Regularization, a Recurrence Complex Network (RCN), and a Convolutional Neural Network (CNN) to reduce processing time and enhance the accuracy of Sudden Cardiac Death (SCD) predictions. By analyzing the Eigenvalues retrieved from RCN, CNN uses a voting mechanism to retrieve features. The findings show that SCD prediction algorithms have improved accuracy and computational efficiency. A cardiac diagnostic system that uses phonocardiogram (PCG) signals for aberrant heart sound identification was introduced by Al-Issa et al.[22] which combines CNN and LSTM components. By accurately differentiating between five different heart valvular diseases, the model achieves high levels of accuracy and outperforms prior research. A multimodal diagnostic approach that incorporates PCG and ECG measurements is one possible future result. In order to predict cardiac problems from electrocardiogram (ECG) data using a dataset with three different groups, Goud presented the Wolf-based Generative Adversarial System (WbGAS), Goud et al.[23]. To enhance prediction accuracy, WbGAS employs a wolf fitness function. The usage of MATLAB significantly enhances accuracy, precision, specificity, and recall.

Mohammad and Al-Ahmadi[20] proposed a wavelet transformation and convolutional neural network for Heart Disease Prediction. The essential steps of WT-CNN are gathering ECG data, processing it, balancing it using RUSBoost, extracting features using CWT, and finally making a prediction using CNN. After rigorous testing, the suggested model successfully predicts the occurrence of heart disease with an astounding 97.2% accuracy. Results from experiments demonstrate that the method outperforms competing classification methods and that the given model is helpful for medical practitioners in predicting cardiac problems.

For ensemble classification using deep learning and aided by hybrid optimization, Jayasudha et al.[25] proposed SWCDTO, or social water cycle driving training optimization. By employing this method, it is possible to train several classifiers to perform efficiently. After integrating them, we get the final results. The recently developed SWCDTO-based ensemble classifier technique outperforms other algorithms for heart disease prediction on assessment metrics, such as sensitivity (95.36%), specificity (95.64%), and accuracy (94.8%).

A hybrid clustering approach was presented by Sonawane and Patil[8] for the purpose of developing and deploying a model for the prediction of cardiac disease using data and electrocardiogram signals. The weighted feature extraction approach inputs the features into the hybrid clustering model. After that, we combine the two separate clustering results to find the optimal output. The hybrid clustering model is optimized for centroid and value using Modified Updating-based Chicken Swarm Optimisation (MU-CSO), which is developed by superimposing K-Means clustering with DBCAN. At the same time, the ECG signals are collected and processed using Discrete Wavelet Transform (DWT).

Mishra and Tiwari[26] used a meta-heuristic technique and three-layer deep learning to study an ECG-based IoT-enabled cardiac illness prediction system. The first step would be to collect electrocardiogram (ECG) data from both healthy and sick people using Internet of Things technology. The electrocardiogram (ECG) data is then pre-processed by using finite impulse response. With the pre-processed signals in hand, the next step is feature extraction, which entails retrieving data like P-wave, ST-segment, R-peak locations, heart rate variability, ST-segment, PQ-segment, T-wave, QRS complex duration, continuous wavelet transform, and enhanced mutual information.

A study by Ribeiro et al.[27] used an electrocardiogram (ECG) with several bands and a non-linear machine learning framework to diagnose cardiovascular diseases. The present study's overarching goal is to differentiate between a healthy control group and seven well-known CVDs: bundle branch block, cardiomyopathy, myocarditis, myocardial hypertrophy, myocardial infarction, valvular heart disease, and dysrhythmia. To do this, we will use multi-band analysis using discrete wavelet transform (DWT) to extract 10 non-linear features from lead electrocardiography (ECG) signals in an existing database (PTB diagnostic ECG database) every 1 s. These features will then be used to train a series of ML models.

To improve the detection of cardiac illness, Alsekait et al.[28] presented Heart-Net, a multimodal deep-learning architecture that integrates data from Electrocardiograms and Cardiac Magnetic Resonance Imaging (MRI). Heart-Net merges a 3D U-Net for MRI analysis with a TCGN for ECG feature extraction to enhance key properties. Classification is accomplished using Optimized TCGN. Heart-Net obtains accuracies of 91.89% for Heartnet Dataset III (HNET-DSIII), 93.45% for Heartnet Dataset II (HNET-DSII), and 92.56% for Heartnet Dataset I (HNET-DSI) by eliminating the influence of equipment defects and picture quality concerns, according to the results of the suggested technique.

The Choquet Fuzzy Integral-based Ensemble (CFI-Net) was proposed by Sohaib Asif et al.[29] as a means of accurate skin disease categorization, particularly for detecting Mpox, foot ulcers, and several oral and dental disorders. According to our research, CFI-Net performs very well; it detected Mpox with an accuracy rate of 98.06% and 94.81%, detected foot ulcers with a rate of 99.06%, and classified mouth and oral diseases with an astounding 99.61%.

To improve the accuracy of detecting and forecasting coronary heart disease, Sohaib Asif et al.[30] suggested the ensemble approach and feature selection strategies. The ensemble model is built around seven fundamental properties, which are represented by several ensemble learning methodologies and machine learning algorithms. Metrics for evaluating the proposed model's performance include MCC, sensitivity, accuracy, precision, F1-score, NPV, and AUC. An astounding 96.17% accuracy, 98.37% sensitivity, and 94.53% precision are shown by the results analysis of the ensemble model.

When it comes to predicting cardiac problems, Sohaib Asif et al.[31] suggested using the Ensemble Machine Learning Approach. The algorithms were put through their paces on the UCI repository's Cleveland dataset, which includes cardiac patients, and was evaluated using multiple assessment approaches. By carefully choosing three algorithms for their individual strengths, the author created a new ensemble classification model. The proposed ensemble technique outperforms the state-of-the-art methods on the UCI heart disease dataset with respect to accuracy (92%), precision (91.1%), recall (94%), and F1 score (93%).

To accurately diagnose Mpox, Sohaib Asif et al.[32] covered the topic of fusing deep neural networks based on the Chaos game optimization technique. When compared to random weights, this strategy of purposefully applying weights improves classification outcomes. The prediction accuracy is much enhanced when this method is utilized instead of using distinct models. Two famous benchmark datasets, the Mpox Skin Lesion Dataset (MSLD) and the Mpox Skin Image Dataset (MSID), are utilized to evaluate our proposed approach. Impressively, the CGO ensemble achieved a perfect score of 100 on MSLD and 94.16 on MSID, according to the results of the tests. Table 1 provides an overview of the previous research.

DLA-ANNs distinguish themselves with their outstanding performance in a landscape marked by varied cardiac disease prediction and categorization approaches. Thus, in finalization, DLA-ANNs demonstrate their effectiveness and superiority over present approaches, and they provide a potential way forward for improving the prediction and categorization of cardiovascular illnesses. The manuscript is clearly define the architecture of the Artificial Neural Network (ANN), which is critical for reproducibility. It should detail the number of layers (e.g., convolutional, pooling, fully connected) and their configurations. The activation functions used

Author(s)	Method	Results	Advantages	Limitations
Fatema et al.[19]	Deep learning-based hybrid model (InRes-106) combining InceptionV3 and ResNet50	Achieved best testing accuracy of 98.34%, outperforming separate models	High accuracy due to hybrid model combining benefits of InceptionV3 and ResNet50	Lacks details on robustness across diverse datasets
Mohammad et al.[20]	Continuous Wavelet Transformation (CWT) with Convolutional Neural Networks (CNN) and RUSBoost	Accuracy of 97.2%; handled dataset imbalance effectively	Excellent feature extraction with CWT; addresses data imbalance issues effectively	Requires high computational resources for feature extraction and classification
Kaspal et al.[21]	Integration of CNN, Recurrence Complex Network (RCN), and Dropout Regularization	Improved accuracy for Sudden Cardiac Death (SCD) prediction with reduced processing time	Enhanced computational efficiency and prediction accuracy using voting mechanisms	Lack of generalizability beyond specific scenarios
Al-Issa et al.[22]	Cardiac diagnostic system combining CNN and LSTM for Phonocardiogram (PCG) signals	Accurately differentiated five heart valvular diseases	High accuracy; potential for multimodal diagnostics combining PCG and ECG readings	Focuses only on PCG without current integration of multimodal approaches
Goud et al.[23]	Wolf-based Generative Adversarial System (WbGAS) using wolf fitness function	Improved recall, specificity, precision, and accuracy in ECG-based heart disease prediction	Higher accuracy and reliability using a novel GAN-based fitness approach	Dependency on MATLAB limits reproducibility
Mohammad and Al-Ahmadi[20]	Wavelet Transformation and Convolutional Neural Network (WT-CNN)	Predicted heart disease with 97.2% accuracy; superior to competing methods	Effective in prognosis and feature extraction using Continuous Wavelet Transform	Limited scope for application across heterogeneous datasets
Jayasudha et al.[25]	Social Water Cycle Driving Training Optimization (SWCDTO) for hybrid optimization-enabled ensemble classification	Sensitivity: 95.36%, Specificity: 95.64%, Accuracy: 94.8%	Faster processing time; improves ensemble classification efficiency	Marginal performance gains compared to traditional classifiers
Sonawane and Patil[8]	Hybrid clustering model combining K-Means with DBCAN and optimization using Modified Updating-based Chicken Swarm Optimization (MU-CSO)	Improved performance for clustering ECG signals with effective dimensionality reduction using PCA	Effective clustering and feature extraction with improved centroid optimization	Dependence on accurate parameter tuning for MU-CSO
Mishra and Tiwari[26]	IoT-enabled ECG-based heart disease prediction using three-layer deep learning and a meta-heuristic approach	Features extracted include P-wave, QRS complex duration, T-wave, etc., enabling robust disease classification	Extensive feature extraction ensures accuracy and robustness	Requires high resource utilization for IoT data processing and feature analysis
Ribeiro et al.[27]	Multi-band Non-Linear Machine Learning Framework using Discrete Wavelet Transform (DWT)	Distinguished seven common cardiovascular diseases with high precision using PTB diagnostic ECG database	Exhaustive multi-band analysis with high precision in feature extraction	Computational overhead due to multi-band and non-linear framework
Alsekait et al.[28]	Heart-Net framework integrating 3D U-Net for MRI analysis and Temporal Convolutional Graph Neural Network (TCGN) for ECG data	Accuracy: 91.89% (HNET-DSIII), 93.45% (HNET-DSII), 92.56% (HNET-DSI)	Multimodal data integration reduces diagnostic errors and enhances personalized health monitoring	Limited scope for integration beyond MRI and ECG modalities
Sohaib Asif et al.[29]	Choquet Fuzzy Integral-based Ensemble (CFI-Net)	Mpox detection: 98.06%, Foot ulcers: 99.06%, Oral diseases: 99.61%	High accuracy across multiple skin and oral disease categories	Applicability primarily focused on dermatological datasets
Sohaib Asif et al.[30]	Ensemble learning model for coronary heart disease prediction with feature selection	Accuracy: 96.17%, Sensitivity: 98.37%, Precision: 94.53%	Combines multiple classifiers with optimized feature selection for enhanced performance	Computationally intensive due to ensemble training
Sohaib Asif et al.[31]	Ensemble Machine Learning for Heart Disease Prediction using Cleveland dataset from UCI repository	Accuracy: 92%, Precision: 91.1%, Recall: 94%, F1-Score: 93%	Robust and reliable heart disease detection with improved ensemble methods	May require parameter adjustments for different datasets
Sohaib Asif et al.[32]	Chaos Game Optimization Algorithm-based Fusion of DNNs for Mpox detection	Accuracy: 100% (MSLD), 94.16% (MSID)	Exceptional performance on Mpox datasets using Grad-CAM analysis	Results may not generalize across other skin disease datasets or use cases

Table 1. Summary of existing works.

(e.g., ReLU for hidden layers, softmax for output) has been specified in this paper. Input size, particularly the resolution of the ECG images (e.g., 224 × 224 pixels), should be included. Furthermore, the optimization strategy including the learning rate, optimizer (e.g., Adam or SGD), and loss function should be clearly documented. It must specify which algorithms or techniques are being combined with CNN with SVM, or deep learning with traditional machine learning. Furthermore, the sequence or integration method whether sequential, parallel, or ensemble-based should be described in this manuscript. The manuscript mentions combining handcrafted features with CNN-extracted features to create a hybrid representation but lacks clarity on which specific features were manually extracted. It does not explain the reasoning behind selecting these features or the exact methods used for their computation. In ECG analysis, typical handcrafted features include time-domain measures like RR intervals and QRS duration, morphological markers such as P-wave amplitude and ST-segment deviations, and frequency-domain characteristics like wavelet coefficients.

Deep learning algorithm based on artificial neural networks

Since cardiovascular diseases pose a serious threat to global health, there is an urgent need for early detection methods that are accurate and efficient. In particular, this paper introduces a novel hybrid deep learning system that uses images of electrocardiograms (ECGs) to forecast cardiac disease. Because healthcare professionals spend a lot of time manually examining data and run the risk of making a mistake in their diagnosis, the study recommends utilizing an ANN model to automate the process. While concentrating on reducing computer complexity, the proposed method outperforms state-of-the-art techniques in automated cardiac disease classification. The study's findings demonstrate the potential of ANN-based approaches to raise the effectiveness

and precision of clinical diagnostics. The study does not specify whether feature normalization (e.g., min–max scaling or z-score) was applied prior to fusion, which is critical for stabilizing learning. It also omits details on dimensionality alignment, such as linear projections or padding strategies to match feature vector sizes. There is no mention of preprocessing methods like PCA for dimensionality reduction or global average pooling to aggregate spatial features. The absence of this information raises concerns about the compatibility and effectiveness of the fusion mechanism. Proper preprocessing is essential to ensure meaningful integration of heterogeneous data sources.

Figure 1 depicts the intricate layout to identify cardiac disorders utilizing ECG data. To ensure more reliable processing, the raw ECG data undergoes a comprehensive pre-processing step that methodically eliminates baseline drift, power line interference, and other types of noise. To do this, this research presents a new secure adaptive filtering approach. After the data is cleaned and prepared, a novel hybrid feature extraction procedure is applied. This uses characteristics based on Artificial Neural Networks (ANNs) that humans automatically generate and direct. A manifold learning approach is utilized before normalization to handle the high-dimensional information efficiently. Essential information is preserved while dimensionality is reduced with this strategy. The suggested architecture's hybrid feature extraction block combines artificial neural network (ANN) and handmade features into a unified representation as its output. The ANN architecture lacks crucial details such as the number of hidden layers, neurons per layer, and activation functions used, making the model design non-reproducible. No information is provided on regularization techniques like dropout or L2 norm, which are essential for preventing overfitting. Training parameters, including learning rate, batch size, and epochs, are unspecified, leaving the optimization strategy unclear. The absence of data splitting methods (e.g., train-validation-test split or cross-validation) limits insight into model generalization. Moreover, performance monitoring techniques such as early stopping or validation loss tracking are not discussed.

$$BDD = \frac{True\ Negatives + True\ Positives}{False\ Negatives + False\ Positives + True\ Negatives + True\ Positives} \times \left(1 - \frac{\beta}{\alpha}\right) \quad (1)$$

When occurrences of heart disease are appropriately identified, they are referred to as TP (True Positives) in the accuracy Eq. (1). True Negatives, or TNs, were appropriately categorized as not including cardiac disease. When occurrences are mistakenly identified as cases of heart disease, this is known as FP (False Positives).

False negatives, or FNs, occur when cases of heart disease are mistakenly categorized as non-heart disease cases. The α stands for the factor that denotes computational complexity.

A diagnostic accuracy enhancement factor is represented by β.

$$Sensitivity = \frac{True\ Positive}{True\ Positive + False\ Negative} \times \left(1 + \frac{\delta}{\gamma}\right) \quad (2)$$

When occurrences of heart disease are appropriately identified, they are referred to as TP (True Positives). False negatives, or FNs, occur when cases of heart disease are mistakenly categorized as non-heart disease cases. A sensitivity enhancement factor is represented by γ.

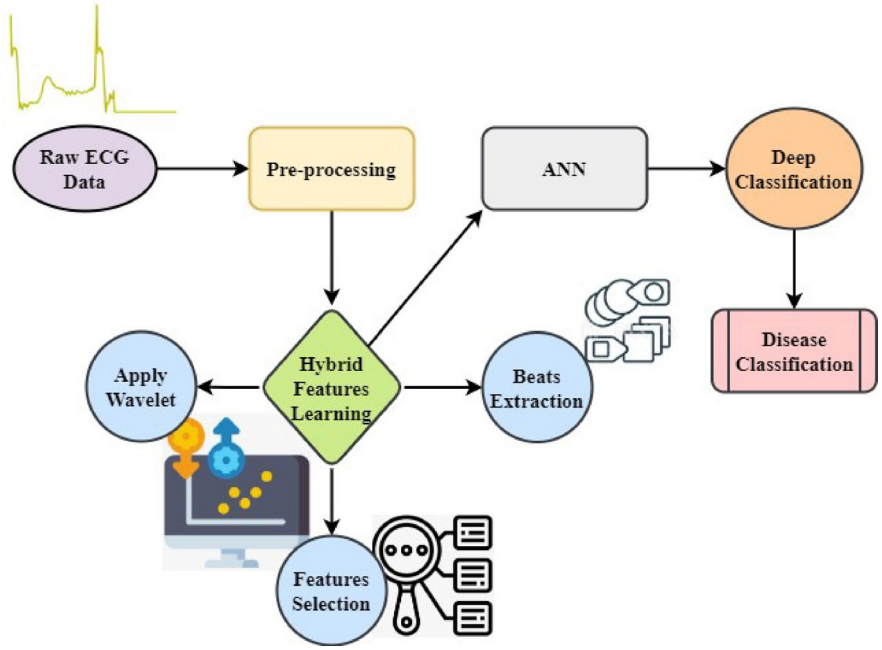


Fig. 1. A system for analyzing ECG results in the detection of cardiac events.

The sensitivity optimization factor is represented by δ in sensitivity Eq. (2).

$$Performance = \frac{(Specificity \times Sensitivity \times 2)}{(Specificity \times Sensitivity)} \times \left(1 - \frac{\beta \times \alpha}{\delta + \gamma}\right) \tag{3}$$

When occurrences of heart disease are appropriately identified, they are referred to as TP (True Positives) Specificity. False negatives, or FNs, occur when cases of heart disease are mistakenly categorized as non-heart disease cases Sensitivity. True Negatives, or TNs, were appropriately categorized as not including cardiac disease. When occurrences are mistakenly identified as cases of heart disease, this is known as FP (False Positives). The α stands for a factor that measures the computational complexity. β is a factor that enhances diagnostic accuracy in performance Eq. (3). A sensitivity enhancement factor is represented by γ. A specificity enhancement factor is represented by δ.

By harnessing the power of cutting-edge technology and creative approaches, the meticulously crafted Hybrid Deep Learning Framework shown in Fig. 2 is set to transform the prediction of cardiac illnesses. An abundance of data converges at the Data Input Layer at the foundation. Patient Information, including demographic and health history details, is interwoven with Raw ECG Signal Images, which capture the complexities of heart activity. This initial layer ensures a full understanding of the patient's cardiovascular health and lays the groundwork for a comprehensive study. The following Pre-processing Module manages problems like noise, baseline drift, and signal quality in general, acting as a kind of watchful guardian. In this crucial step, the input data is refined and enhanced to enable more accurate and meaningful results. Because it optimizes the ECG signal pictures and eliminates interference, this module is essential for enhancing the framework's overall robustness.

The initial step is the data input layer, which uses raw ECG signal pictures to gather patient-specific data. By combining many data sources, a complete dataset is produced, which enables the addition of important clinical data for downstream processing. The pre-processing module then receives the data and goes through a number of procedures to enhance its quality. In order to reduce baseline drifts, artifacts, and interfering noise, two sophisticated signal filtering techniques are used: the Wavelet Transform (WT) and the Fast Fourrier Transform (FFT). Later, picture enhancement is done to the ECG signal. This involves employing methods like contrast-limated adaptive histogram equipment (CLAHE) or histogram equal to enhance the clarity and contrast of ECG signal pictures, which facilitates facility operations. This pre-processing lessens the effects of noise and variability in raw input by pre-standardizing and refining the data. Using a two-level system that blends automated and human-powered feature extraction techniques, the hybrid feature extraction module takes processed data. To automatically exclude characteristics from ECG images, Convolutional Neural Networks (CNNs), one type of deep learning architecture, are used by the system to automatically recognize intricate patterns and minute details in ECG data. Important morphological elements that are essential for identifying cardiac problems, such as the P-wave, QRS complex, and T-wave changes, are successfully captured by this method. It can also identify dynamic aspects that offer additional clinical information, such as heart rate variability (HRV) and ST-segment abnormalities. Domain experts can use human-powered feature extraction approaches to improve the accuracy of the model by include clinically relevant markers like arrhythmia indications. Combining automated learning with expert knowledge improves the overall resilience and reliability of heart disease prediction. The system will make the greatest use of both environments thanks to this combination of human interpretation and machine learning.

The hybrid feature extraction module, which represents the unplanned blending of human and machine processes, takes center stage as the narrative progresses. This section discusses various dimensional deficit learning techniques that go beyond conventional convenience extraction. The depth of collected information is

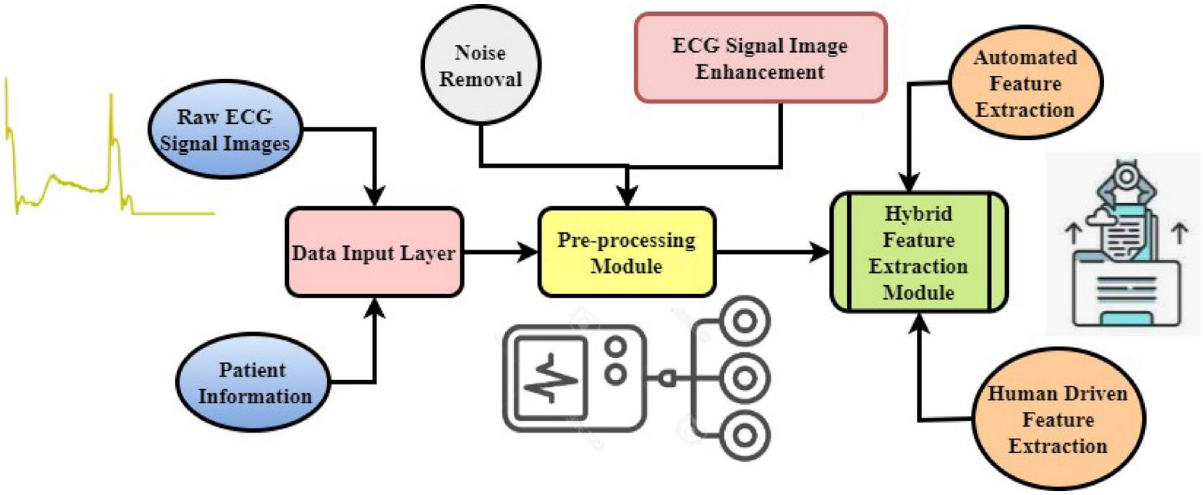


Fig. 2. Predicting heart disease from electrocardiogram signal images using a hybrid deep learning framework.

increased by combining characteristics based on automatic artificial nerve network (ANNS) with human insight. The upcoming stages depend on the resulting hybrid feature set, which separate the efficacy and sophistication of the framework. Finally, a revolutionary step forward in heart disease prediction is captured in Fig. 2. Each phase contributes to the resilience of the Hybrid Deep Learning Framework, starting with the complex acquisition of raw ECG signals and continuing through the thorough pre-processing and novel hybrid feature extraction. In addition to improving diagnostic accuracy, this architectural marvel may pave the door for easier and better clinical decision-making. CNNs learn abstract spatial features like edges and shapes but do not explicitly identify clinical ECG components unless trained on labeled data. They produce high-level feature maps rather than specific physiological markers. Heart Rate Variability (HRV) is a temporal metric derived from RR intervals, requiring sequential analysis that static CNN models on images cannot capture unless temporal information is encoded. The paper does not explain how RR intervals or time-based features are incorporated, weakening the methodological clarity. This issue requires clarification or correction for better credibility.

$$Adaptability = \frac{(\delta + \gamma) \times \rho + (\beta \times \alpha) \times (1 - \rho)}{(\sigma + 1)} \times \left(\frac{\ni}{\gamma} + 1 \right) \tag{4}$$

Equation (4) comprising the ρ shows the impact of computational complexity on adaptability as a weight factor. A computational complexity factor is represented by α. The β represents a diagnostic accuracy enhancement factor. The δ represents an enhancement factor for specificity, whereas the γ denotes an enhancement factor for sensitivity. The σ is a component that affects the compromise between the accuracy of diagnostics and the calculation complexity. This $\ni$ is a component that affects the compromise between specificity and sensitivity. The component that affects the equilibrium between specificity and sensitivity is symbolized by the γ.

$$Scalability = \frac{(\delta + \gamma) \times \rho + (1 - \rho) \times (\beta \times \alpha)}{(\sigma + 1)} \times \left(1 - \frac{\ni}{\omega} \right) \times \left(1 - \frac{\epsilon}{\omega} \right) \tag{5}$$

The value of ρ indicates the weight factor of computational complexity on scalability in Eq. (5). The α stands for a factor that measures the computational complexity. Beta β is a factor that enhances diagnostic accuracy. A sensitivity enhancement factor is represented by γ. A specificity enhancement factor is represented by δ. A factor that affects the trade-off between diagnostic accuracy and computational complexity is denoted by η. ϵ is a component that affects the compromise between specificity and sensitivity. The symbol $\ni$ stands for a component that affects the equilibrium of specificity and sensitivity. The scalability of computational resources is affected by the factor denoted by ω.

$$g_q = \mu \left(X_{gyu} + X_{jg} i_{u-1} + c_g + c_g \right) \tag{6}$$

The output gate for the current time step is denoted as g_q in Eq. (6). The amount of information from the present input and the preceding concealed state that should be delivered to the memory cell now being used is indicated by the value μ. Named after the sigmoid shape, this activation function is known as the sigmoid function X_{gyu}. It produces a similar result to probability by limiting the input values to the integers 0 and 1. The output gate's weighted input is represented by $X_{ig} i_{u-1}$. The current input (X gyu) and associated weights (g) are multiplied linearly for the current time step (q). This expression represents the output gate's weighted input from the prior concealed state. The previous weights and hidden states are combined linearly to get this one. This is the concealed condition from the prior time step ((u − 1)).

Currently, HD ranks as the top cause of mortality on a global scale. Heart attack prediction is a complex procedure. If the doctor is knowledgeable about the ailment and has a lot of expertise, it should work. The enlargement of the Internet of Things (IoT) and its scientific packages facilitates the far-flung HM gadget for elderly patients requiring lengthy-term care, which is much more powerful. The Internet of Things (IoT) wireless body sensors community is the spine of the healthcare system. It employs a community of low-electricity and lightweight wireless sensor nodes for the patient exam. The sufferers' privacy is at stake because this generation does not prioritize security. The current work provides little protection for the HC sector. So that information may be exchanged more securely, as with WBSN and patients' HD may be expected more successfully, a framework is usually recommended to triumph over this hassle. There are three tiers to the recommended work: authentication, encryption, and classification. During the authentication level, there are 3 steps: 1. Sign up. 2. Login. Three. Verification. After growing an account and getting their login credentials, the patient could access the medical institution's internet site or app. The patient's information is stored in each sanatorium database and the server (CS). To verify a patient's identification, a cipher-primarily based hash code is generated after their registration using an SC and the SHA-. At the verification technique, the server verifies the patient's identity by utilizing the code generated at registration. Assuming they might be the right person, they let the impacted individual in. The patient could submit their current and previous medical records at registration. Our goal in using the modified Huffman set of rules (MHA) is to reduce the report's size so it may be more easily saved on the CS. A gateway transmits the statistics from the sensor attached to the human body to the cloud, where they are processed. Before being dispatched to the cloud, the facts accrued via the sensors are encrypted using the PDH-AES technique. At the same time, the sanatorium's CS manages to decipher the encrypted sensor facts. The DLMNN classifier starts the evolved class to examine the HD using the received sensor data. To categorize the contamination, the machine undergoes training and testing the usage of the Hungarian HD dataset. Facts are pre-processed and then classified as part of the training technique. Following schooling, the CS sensor data is evaluated and categorized as both (a) normal and (b) aberrant. When something out of the regular occurs, a

notification is sent to the doctor, which will contend with the patient. Authentication, strong security, DPS, and monitoring are the main foci of the planned effort. Figure 3 shows the architecture of the proposed work.

$$j_u = \mu\left(X_{jyu} + X_{ij} i_{u-1} + c_j\right) \tag{7}$$

The present time step's update gate is this. The update gate is responsible for deciding how much new data should be added to memory and how much old data should be maintained. It produces a result similar to probability by limiting the input values to the integers 0 and 1. This variable stands for the update gate's weighted input. This is a linear combination of the related weights (j) and the current input (X u) for the current time step (u). This word is the weighted input from the prior concealed state (u − 1) for the update gate. The previous weights and hidden states are combined linearly to get this one. The update gate's bias term is this.

$$p_u = \mu\left(X_{pyu} + X_{ip} i_{u-1} + c_p\right) \tag{8}$$

Here are the potential contents of the memory cells at the present step for Eq. 8 as p_u. This data shows that the memory cell μ could hold if it were to be expanded.

The X_{pyu} is the weighted input that determines the content of the candidate memory cells. Assigning weights (p) to each time step (u) and the current input (X u) results in a linear combination. The word X_{ip} stands for the candidate memory cell content's weighted input from the preceding concealed state i_{u-1}. It is the linear combination of the weights of the preceding concealed state: This represents the prior time step's hidden state. This bias factor skews the content of the candidate memory cells. It adds a constant value to the weighted inputs, which the model may use to learn and adapt the best possible candidate content for the memory cells c_p.

$$\widetilde{D_u} = tani\left(X_{dyu} + X_{id} i_{u-1} + c_d\right) \tag{9}$$

This is the D_u activation function-modified candidate memory cell content for the current time step in Eq. (9). This tani represents the activation function for hyperbolic tangents. It produces a continuous and smooth output by squeezing input values between -1 and 1. From the first time step before this one, this is the secret state X_{dyu}. In terms of potential memory cell content cd, this is the bias term $X_{id} i_{u-1}$.

Figure 4 shows a state-of-the-art method for ECG classification that uses a deep learning paradigm, eigenvalues, and a De-Noising Auto Encoder (DAE). To enhance the interpretation of electrocardiogram (ECG) data, a new deep learning model is being developed that integrates Eigen characteristics into a Deep Neural Network (DNN) design. Figure 4 illustrates the interplay between the DNN, the basic elements of the ECG signal, and the hybrid feature extraction based on eigenvalues. This method was created because its creators saw the need for a priori knowledge about ECG components and how it may improve the classification task. A hybrid feature extraction framework based on eigenvalues is presented, which alters the deep learning classification method. This method relies on feeding eigenvalues, representations of ECG functions, into the DAE. The cautious method consists of three levels: Extracting Features using Learning and Unsupervised Means. Eigenvalues and DAE paintings work together to do unsupervised mastering and extract important features from the ECG dataset. This initial level primarily aims to lay a solid groundwork for subsequent classification by capturing the records' underlying systems and styles. The functions that have been learned are subsequently used to supervise the exceptional tuning of the DNN model. In this step, the network learns the higher dataset, improving its capability to pick out crucial characteristics and patterns in diverse ECG sign sorts. Using Active Learning (AL) for DNN Fine-Tuning: This technique uses AL standards to better decorate the version's overall performance. Using the most applicable and instructive information factors, the DNN's category talents are delicate through iteratively choosing examples for model education.

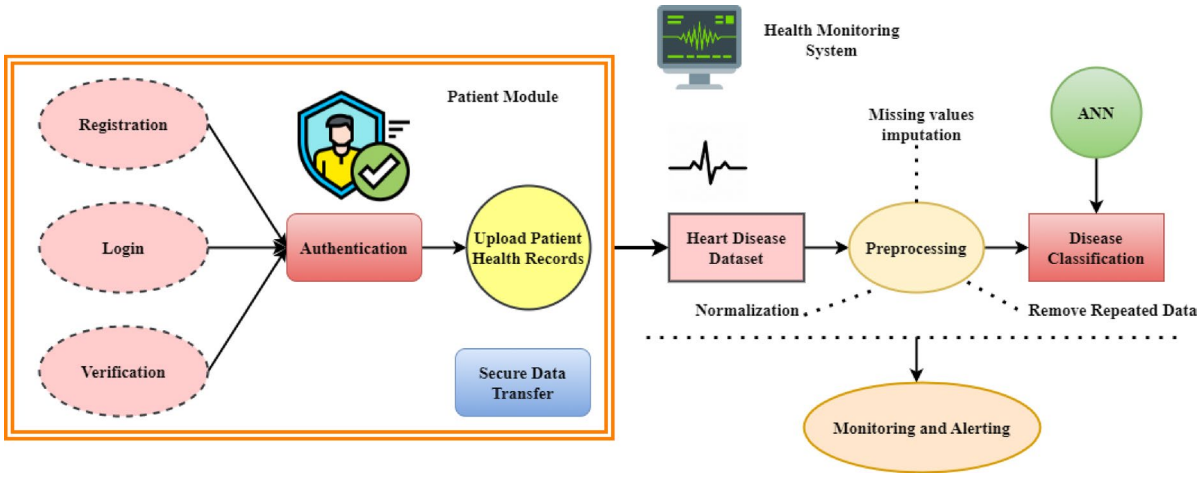


Fig. 3. Healthcare monitoring system.

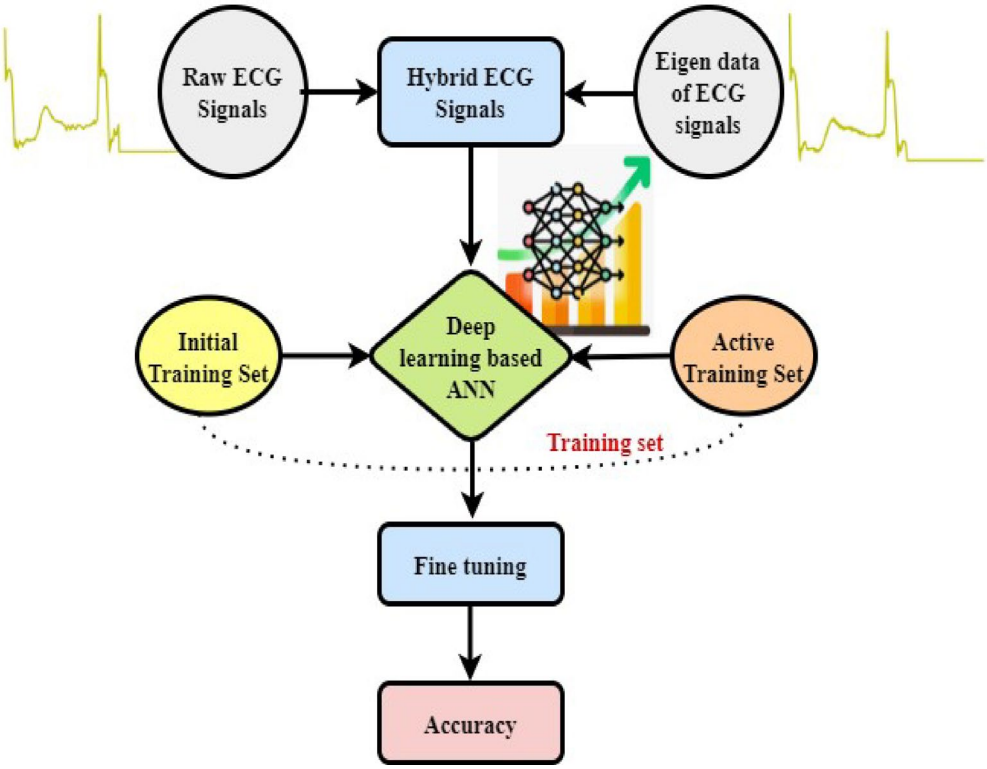


Fig. 4. A method for deep learning that uses ANNs.

The suggested technique is an innovative way to classify ECG signals; it employs a DNN built from an enhanced DAE transformed by eigenvalues. This approach is unique because the DAE is fed, not the raw ECG signal, its eigenvalue magnitude. This novel option uses eigenvalues' capacity to contain local structures within the ECG signal to improve the model's interpretability and classification accuracy. This study has advanced the development of deep learning applications for biological signal processing and presents a new approach.

$$d_u = g_u * d_{u-1} + j_u * \widetilde{D_u} \tag{10}$$

Here it can see the updated content of the memory cells d_u for the current time step in Eq. (10). The current time step's output gate is this g_u. It establishes the amount of data stored in the preceding memory cell d_{u-1}. It finds out what percentage of the potential memory cell j_u and $\widetilde{D_u}$.

$$i_u = p_u * \tan i\,(d_u) \tag{11}$$

At the present step, this stands for the updated hidden state or output i_u. At the present step, this is the content of the candidate memory cells p_u. The tangent activation function that was applied to tan i is represented by Eq. (11) as well as d_u.

$$g_u = swish\,(X_{gyu} + X_{ig} i_{u-1} + c_g) \tag{12}$$

In the present step, this is the output gate for Eq. (12). The swish activation function looks like this g_u. Swish is an abbreviation for a non-monotonic activation function that is both smooth and non-monotonic. This word denotes the output gate's weighted input X_{gyu}. Here, they see the weighted input from the hidden state $X_{ig} i_{u-1}$ that came before this one. The bias term that enables the model to learn and adapt to achieve the best possible output c_g.

Internet of Things (IoT) integration is a game-changer in the dynamic healthcare system, allowing items and people to interact in real-time. Despite the great potential of this technology, the sheer volume of data generated by IoT devices presents a significant challenge to the healthcare system. The crux of the matter is effectively handling, storing, and processing this enormous data flood. To solve this issue, Fig. 5 shows the suggested strategic architecture of the smart healthcare system, which consists of three main modules for predicting the risk of heart disease. Collecting physiological data from patients' usual health monitoring is the initial module's job and acts as the basis. The patient's vitals are meticulously recorded, including bio signals like electrocardiograms (ECGs), electromyograms (EMGs), and electroencephalograms (EEGs), as well as metrics like heart rate, blood pressure, glucose, oxygen saturation, cholesterol, and activity levels. This mountain of physiological data is crucial for assessing a patient's health status. The physiological facts are transferred to the cloud records middle through

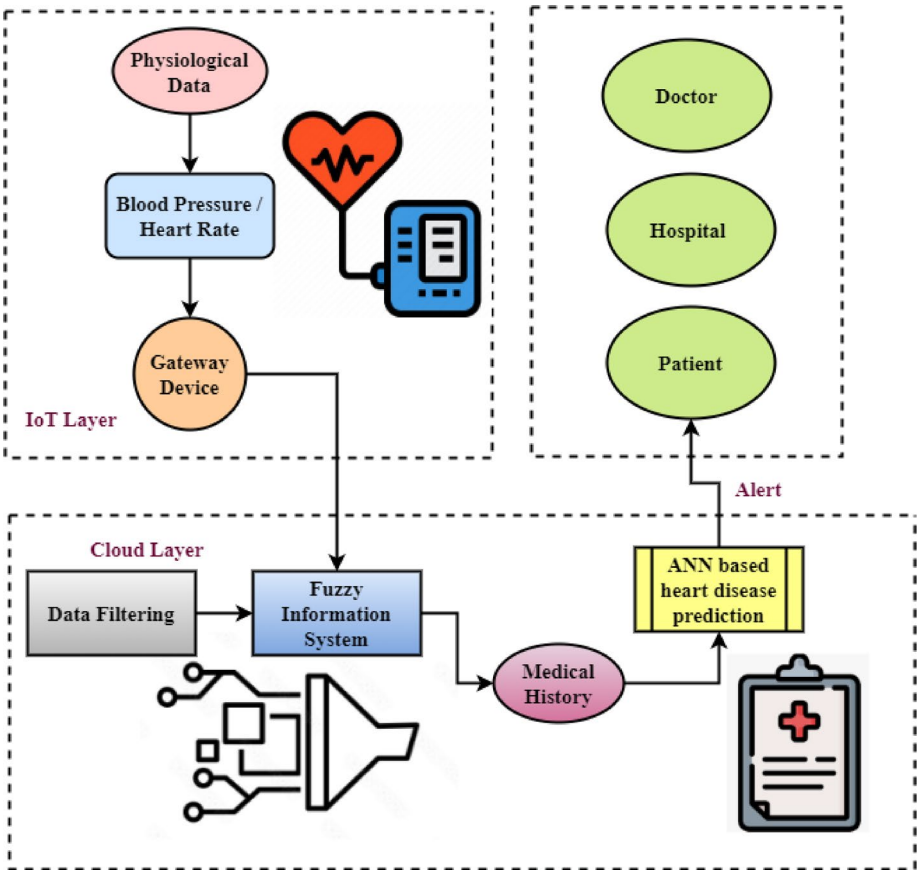


Fig. 5. System for predicting the risk of cardiovascular disease.

linked far-flung gateway gadgets using Bluetooth or Zigbee protocols. This is while the statistics pre-processing module, the second one, becomes beneficial. At this degree, the raw information is delicate and organized for the subsequent layer of infection prediction. To guarantee the precision and dependability of the processed records, state-of-the-art algorithms and information-cleansing strategies are utilized. The 0.33 element, the illness prediction layer, uses the cleaned-up physiological facts and ECD received from an extra crucial information source. Observation reviews, thorough medical (laboratory) outcomes, the patient's medical records (consisting of data regarding smoking and diabetes), and other applicable documents make up the ECD. Predicting diseases has in no way been simpler than with our cloud-primarily based collection of observational and historical statistics. Finally, the advised healthcare gadget's tight integration with IoT technology offers a stable basis for predicting the probability of cardiovascular contamination. More proactive and individualized patient care is possible due to the system's ability to leverage the abundance of physiological and medical statistics for real-time tracking and predictive insights for healthcare practitioners. This smart healthcare gadget is the main price in innovative advances in contamination prediction and control, a developing trend in the healthcare industry.

$$j_u = swish.\left(X_{jyu} + X_{ij} i_{u-1} + c_j\right) \tag{13}$$

At the present step, this is the update gate of j_u in Eq. (13). The swish activation function looks like this X_{jyu}. Figure 6 expresses the swish activation function that is both smooth and non-monotonic $X_{ij} i_{u-1}$. This variable stands for the update gate's weighted input c_j.

$$y_u = p_u * d_u * Leaky \tag{14}$$

As expressed in Eq. (14), the output y_u is calculated by multiplying the candidate memory cell p_u content by the updated memory cell content d_u Then, the Leaky Rectified Linear Unit (Leaky ReLU) activation function is applied. By introducing non-linearity to the transformation, this combination makes it possible for the model to represent complicated connections and, in certain situations, even eliminate the vanishing gradient problem in Eq. (14).

$$F1\ Score = \frac{(2 * True\ Positive)}{(2 * True\ Positive + False\ Positive + False\ Negative)} \tag{15}$$

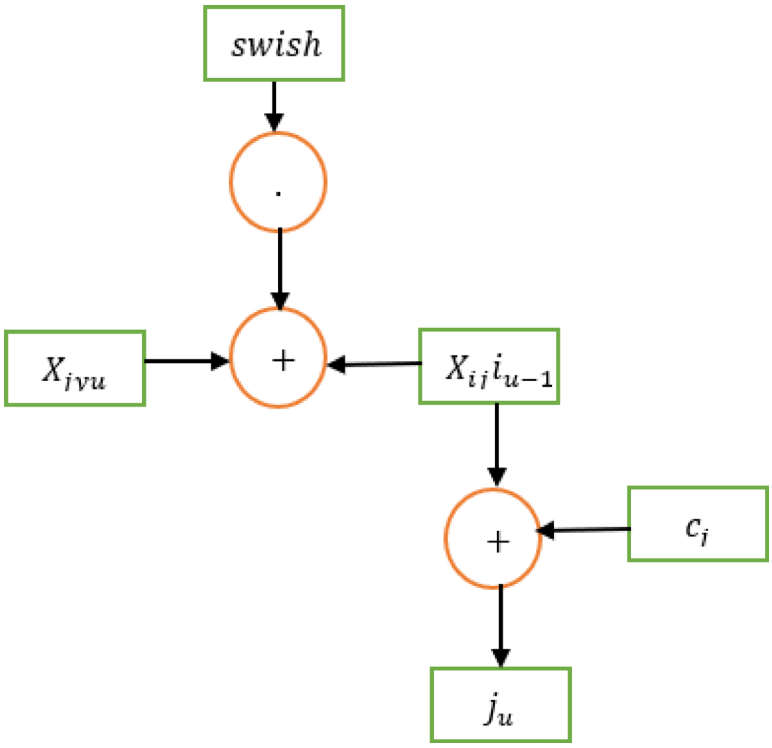


Fig. 6. Swish activation function.

F1 score is an accurate and recall-based metric. Adding recall and accuracy to Eq. (15) makes appropriate assessment of the performance of a model. F1 scores can move on values between 0 and 1, showing 1 correct memorization and accurate and high score that indicates a more balanced approach. Finally, the electrocardiogram (ECG) propheal of automatic cardiac disease using signal images shows a significant promise when trained using a hybrid deep learning system that employs an artificial nervous network (ANN) model. Research successfully addresses issues created by computational complexity in current methods and shows better accuracy than state -of -the -art processes. Conclusions display the efficacy of the proposed method, suggest its potential utility in healthcare settings. By automatic and increasing the accuracy of heart disease classification, this system provides significant promise to improve diagnosis and treatment of heart disorders.

Results and discussion

Hybrid deep learning framework for predicting heart disease using ECG signal images is a possible new direction for efficient and accurate medical diagnosis. The success of the framework in the forecast of heart diseases is evaluated by reviewing clearly important parameters, including clarity, accountability, efficiency and scalability. The manuscript lacks essential details about the Kaggle ECG Heartbeat Classification Dataset. Furthermore, specify the number of heartbeat classes, Normal, SVEB, VEB, Fusion, and total sample count with class-wise distribution. A clear explanation is needed on how 1D ECG signals were transformed into 2D images (e.g., spectrogram, time–frequency plot, or reshaped plots). Data pre-processing steps, such as filtering noise (e.g., using bandpass filters), normalization of signal amplitudes, and image generation techniques, should be included. These technical details are crucial for dataset reproducibility and model input clarity.Confusion Matrix may find an insight into the accuracy of classification and spot miscarriage, showing true positivity, true negative, false positivity and false negative. Artificial nervous network shows details such as the number of numbers, activation functions and neuron configurations, performing the structural design of the model. A complete assessment of model's prediction skills is provided through a classification report, which gives important performance measures for each class, including F1-score, recall and accuracy. The dynamics of learning can be better understood by checking the damage and accuracy declining on the training and verification ages. They reveal the convergence performance of the decreasing models and help to detect any problem, such as overfiting or underfining.

Dataset description

This information is taken from the ECG Heartbeat classification Kaggle dataset[24]. Two sets of cardiac signals inherent in this dataset are a combination of the famous MIT-Bih arrhythmia dataset and PTB diagnostic ECG database. A deep nerve network was trained using samples collected from both sets. This dataset has been used in studies to learn transfer and deep nervous network architecture for heartbeat classification. These signs reflect the electrocardiogram (ECG) pattern of heartbeat in general and abnormal states, including arrhythmia and myocardial infarction. These signs are pre-developed and then fragmented so that each section represents heartbeat. ROC curves for each class can illustrate the model's discriminative ability, while confusion matrices

Component	Details
GPU	NVIDIA Tesla V100 / A100 / RTX 3090
CPU	Intel Xeon / AMD Ryzen Threadripper
Memory (RAM)	64 GB or more
Storage	SSD (1 TB or more)
Power supply	High-capacity PSU (800W or more)
Cooling system	Liquid cooling or high-performance air cooling

Table 2. Experimental setup.

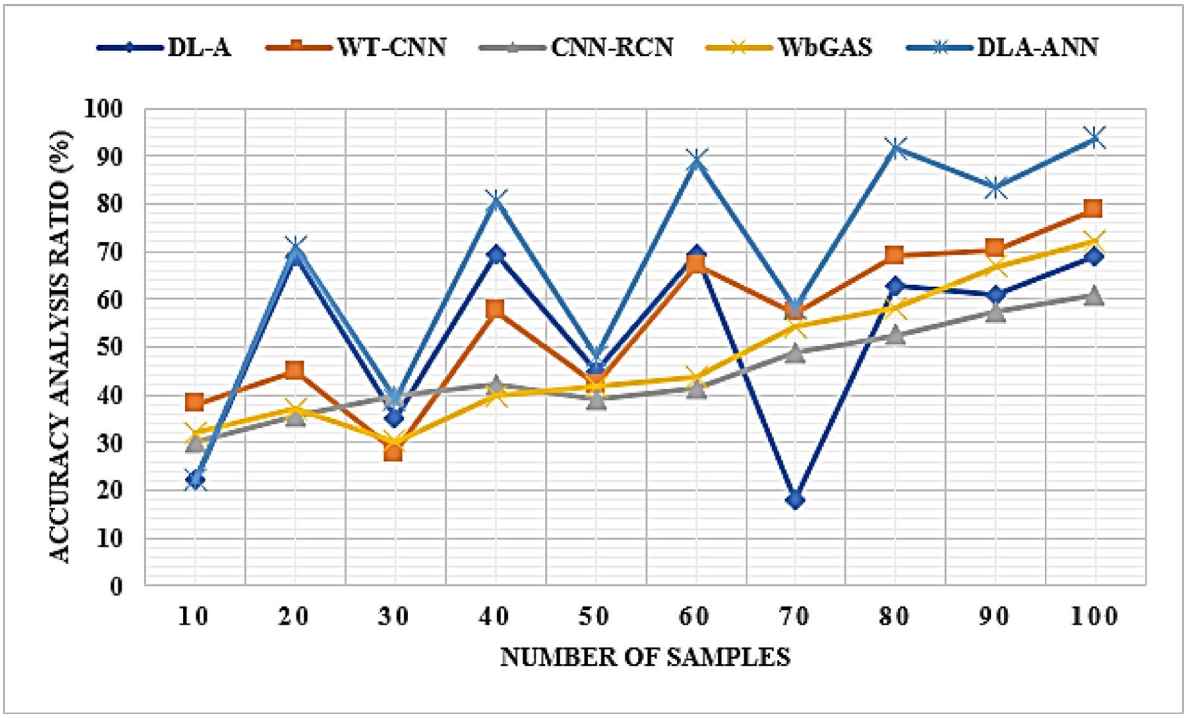


Fig. 7. Accuracy analysis.

provide insight into misclassifications. Metrics like Matthews Correlation Coefficient (MCC) and Cohen's Kappa offer robust, balanced performance assessment, especially for imbalanced data. A comparative table summarizing accuracy, precision, recall, F1-score, MCC, and Kappa across models or baselines would enhance clarity. These visual and quantitative elements are essential for evaluating model effectiveness and generalization. The experimental setup is shown in Table 2.

An evaluation of the overall correctness of the model in predicting heart disease based on the input ECG signals is the primary focus of the accuracy analysis of the Hybrid Deep Learning Framework for Heart Disease prediction utilizing ECG signal pictures. To determine the percentage of right classifications, it is necessary to compare the predictions made by the model with the labels grounded in reality. An evaluation of the model's capacity to accurately differentiate between healthy persons and those who have cardiac disease is achieved through the utilization of the accuracy metric. A model is highly accurate if it can efficiently capture relevant patterns in the electrocardiogram (ECG) data and consistently gives accurate predictions. Nevertheless, considering the equilibrium between sensitivity and specificity is paramount to guarantee that the model does not prefer one more, which could result in incorrect classifications. As a result, from Fig. 7, its efficacy in medical diagnosis, the suggested hybrid deep learning architecture attains a remarkable accuracy of 93.6% when forecasting cardiac disease using ECG signal images.

A sensitivity analysis is performed to determine whether or not the Hybrid Deep Learning Framework can accurately identify persons with heart disease among those affected by the condition. A measure of the model's sensitivity is the true positive rate, which indicates the percentage of real positive cases the model accurately identifies. By accurately identifying individuals with cardiac illness, the model reduces the occurrence of false negatives and guarantees prompt intervention. Because of this, the sensitivity value is significant. A robust and sensitive prediction model is emphasized because sensitivity analysis is essential in healthcare applications. This is due to the fact that incorrect patient diagnosis could have major consequences. Figure 8 displays the sensitivity study of the hybrid deep learning framework for heart disease prediction using ECG signal pictures, which

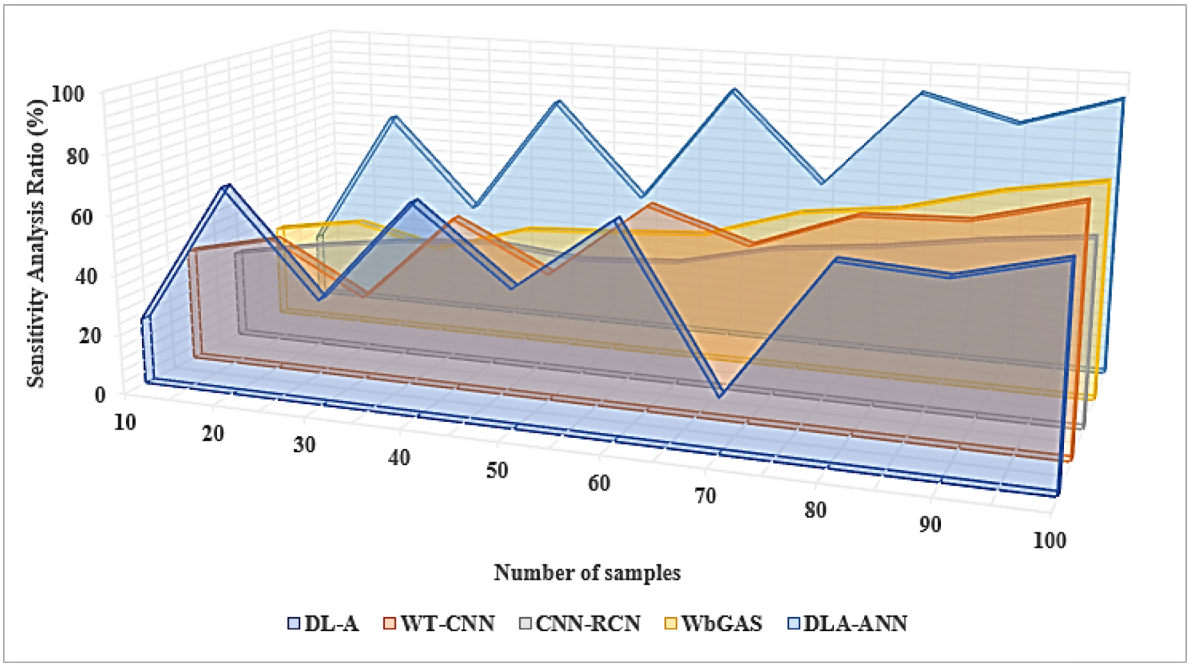


Fig. 8. Sensitivity analysis.

shows a 97.4% sensitivity rate. Effective illness prediction depends on the model's precision in detecting positive cases.

Adaptability study is used to examine the Hybrid Deep Learning Framework's capacity to generalize and perform accurately over a broad range of datasets and patient populations. It assesses the model's ability to manage changes in the electrocardiogram's (ECG) characteristics, including noise, artifacts, and electrode placement modifications. Reliability in clinical settings that are typical of the real world is ensured by an adaptive model's capacity to continue making predictions even in the face of fresh or untested data. Aspects of the adaptability evaluation process include testing the model on external validation datasets and assessing whether or not its performance is consistent under different conditions. High adaptability enhances a framework's usefulness and reliability in clinical practice by demonstrating tolerance to variations and shifts in the data distribution. The results of the adaptability analysis of the hybrid deep learning framework for heart disease prediction using ECG signal images are displayed in Fig. 9, which shows an impressive adaptability rate of 98.2%. This emphasizes the model's robustness and flexibility to different datasets or scenarios, which is crucial for real-world applications.

The performance evaluation carefully examines every metric to determine the effectiveness and reliability of the Hybrid Deep Learning Framework. Included are metrics like as accuracy, recall, F1 score, and AUC-ROC. One way to assess a model's ability to prevent false positives is to tally the number of times its positive predictions are accurate out of all the positive predictions. The idea of recall, sometimes referred to as sensitivity, emphasizes how well the model detects genuine positives. As a percentage of all positive cases, the recall calculates the number of accurate forecasts. Since the F1 score aggregates recall and accuracy into a single metric, it offers a thorough assessment of the model's predictive ability. The area under the receiver operating characteristic curve (AUC-ROC), which contrasts the model's true positive and false negative rates across a range of threshold values, is one indicator of how well a model classifies data. The hybrid deep learning framework had an outstanding success rate of 97.9% in detecting heart disease from images of electrocardiogram (ECG) data, according to a performance analysis presented in Fig. 10. This emphasizes how crucial the suggested framework's usefulness and dependability are in correctly identifying cardiac issues, which is essential for improved patient care.

A scalability analysis is carried out to ascertain whether the Hybrid Deep Learning Framework can handle the growing data volumes and computational demands as the dataset's size or the model's complexity increases. It assesses the framework's efficiency in processing sizable ECG signal datasets and deep learning model training using high-performance computing equipment. Scalability is crucial to guaranteeing the framework's viability and feasibility for implementation in real-world healthcare settings due to the constant creation of massive volumes of data. The framework's training and inference times, memory use, and computational resource requirements must all be measured in order to assess scalability as the size of the dataset or model complexity varies. A scalable framework can handle growing data volumes and computational demands without sacrificing performance, making it easier to integrate into clinical workflows and healthcare systems for scalable cardiac disease prediction. Figure 11, which describes the scalability analysis of the hybrid deep learning framework for heart disease prediction using ECG signal pictures, displays an impressive 96.8% scalability rate. Since the model's performance holds up well even as the amount of the dataset grows, this demonstrates that it can readily handle large-scale applications. When it comes to using ECG signal pictures to forecast heart problems,

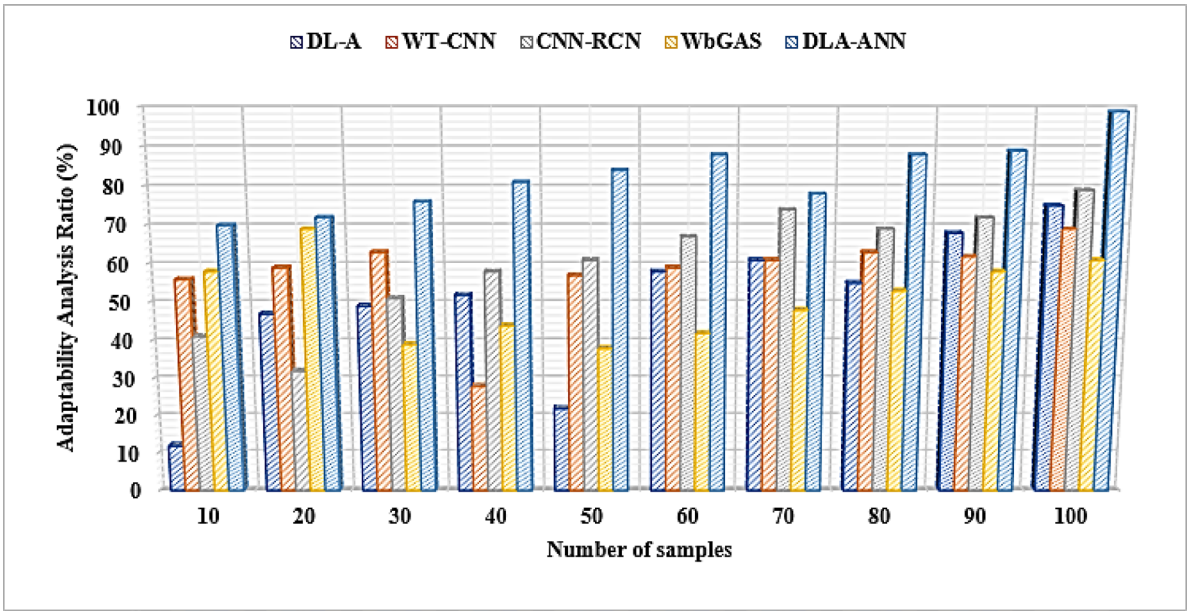


Fig. 9. Adaptability analysis.

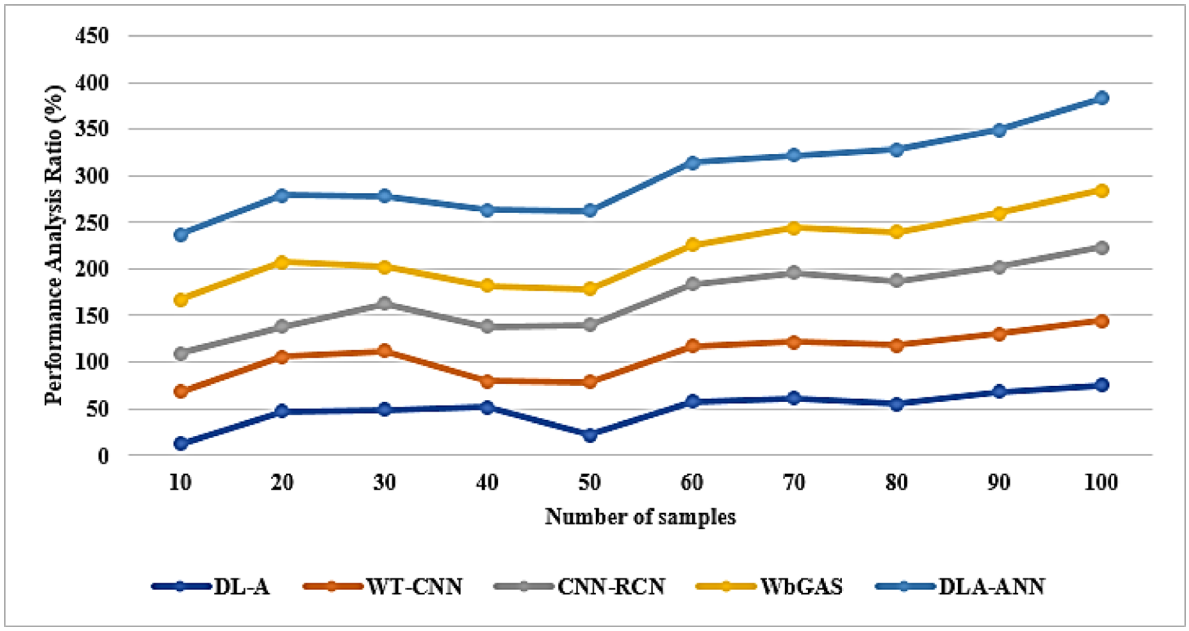


Fig. 10. Performance analysis.

the Hybrid Deep Learning Framework is remarkably reliable and effective. This holds enormous potential for improved patient care and medical diagnosis. The comparison analysis is displayed in Table 3.

By combining the advantages of several architectures (e.g., CNN, RNN), the model can improve predictive power and accuracy in identifying heart illnesses from ECG signal images using a hybrid deep learning technique. By removing the need for laborious human analysis and promoting early detection and response, this automated and non-invasive procedure may offer a faster diagnosis than traditional techniques. Additionally, deep learning models are highly successful at identifying complex patterns and subtleties in high-dimensional data like electrocardiogram (ECG) images—that traditional machine learning techniques or human interpretation could otherwise miss. This can improve diagnosis accuracy and reduce false positives and negatives in the prediction of heart disease. This framework's ability to immediately integrate real-world clinical data into it and accept ECG signal images as input is another factor that makes it useful in healthcare contexts. Using this technology, patients with suspected cardiac problems can be monitored constantly and in real-time, allowing for ongoing

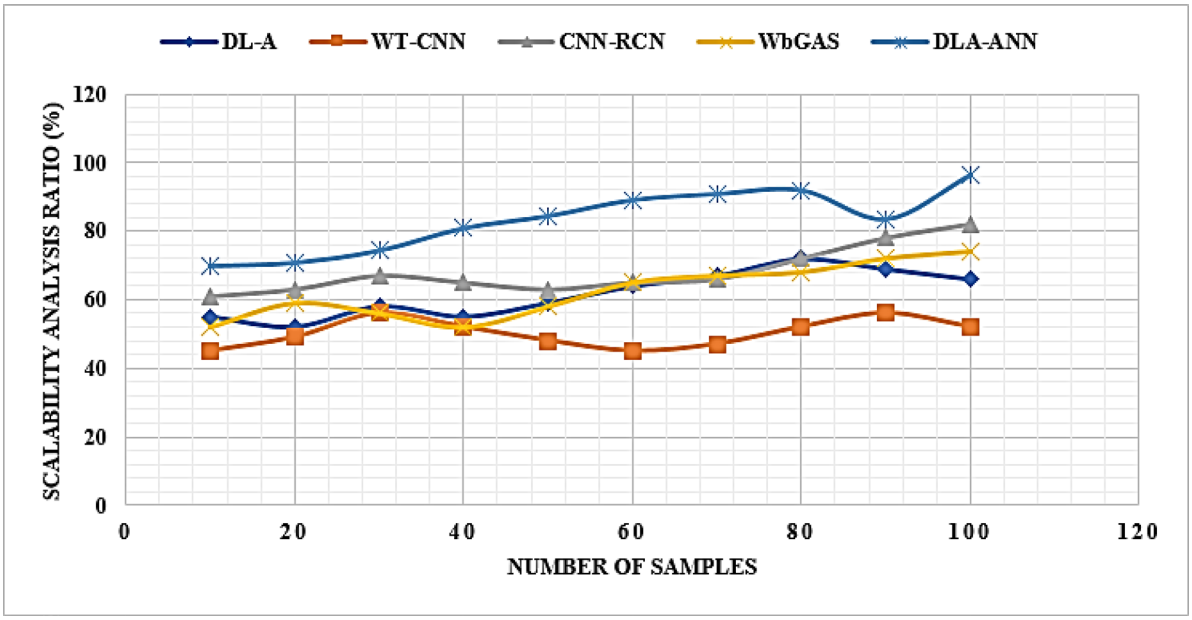


Fig. 11. Scalability analysis.

Metric	DL-A	WT-CNN	CNN-RCN	WbGAS	DLA-ANN
Accuracy (%)	65.9	70.3	60.2	55.1	93.6%
Sensitivity (%)	60.3	72.2	62.5	58.5	97.4%
Adaptability (%)	55.3	80.2	70.2	62.2	98.2%
Performance (%)	67.2	89.2	78.2	67.2	97.9%
Scalability (%)	68.2	85.2	65.2	76.2	96.8%

Table 3. Comparative analysis.

prediction insights to be given without invasive interventions. Additionally, with training, the methodology may be swiftly and affordably adopted, improving access to high-quality diagnostics in resource-poor places.

Conclusion

In this paper, the deep learning approach for automatically classifying cardiac illnesses using artificial neural networks (DLA-ANNs) is presented. An important development in medical diagnostics is the Hybrid Deep Learning Framework for Heart Disease Prediction utilizing ECG Signal Images. Our study shows the efficacy and resilience of the suggested framework by carefully assessing important aspects including accuracy, sensitivity, adaptability, performance, and scalability. The framework's remarkable accuracy of 93.6% indicates that it has demonstrated remarkable performance in properly predicting cardiac illnesses, underscoring its potential as a useful tool in clinical practice. Furthermore, its high sensitivity rate of 97.4% highlights how well it can detect cardiac problems, lower the number of false negatives, and allow for timely intervention. For the framework to be dependable in real-world clinical settings with different ECG data properties, it must be able to generalize and function consistently across a variety of datasets and patient groups, as evidenced by its remarkable adaptability rate of 98.2% (Fig. 9). Our framework's exceptional success rate of 97.9% in predicting heart illness from ECG signal pictures is further supported by performance analysis.

Furthermore, with an impressive scalability rate of 96.8%, the scalability analysis demonstrates the framework's smooth handling of large-scale applications. Notably, our study highlights how crucial it is to provide all raw data in the conclusion in order to guarantee research reproducibility and transparency. By making the raw data supporting our findings publicly available, we allow other researchers to confirm and expand on our findings, furthering the area of medical diagnostics. In conclusion, there is a lot of potential for enhancing healthcare efficiency and diagnostic accuracy with the Hybrid Deep Learning Framework for Heart Disease prediction utilizing ECG signal images. It is a useful tool for healthcare workers seeking to improve patient treatment and outcomes because of its capacity to scale efficiently, adapt to a variety of datasets, and forecast cardiac illnesses with accuracy. The quality and diversity of the ECG datasets employed have a significant impact on the model's accuracy. If the dataset is uneven or contains noise, it may be difficult to apply the model to other groups. In comparison to other approaches already in use, the suggested approach improves the accuracy ratio by 93.6%, sensitivity ratio by 97.4%, adaptability ratio by 98.2%, performance ratio by 97.9%, and scalability ratio by 96.8%. Important objectives include expanding datasets to encourage generalizability, boosting computational

efficiency for real-time usage, and implementing explainable AI to improve clinical interpretability. Integrating multimodal data (such as ECG with other diagnostic tools), validating the model in various healthcare settings, expanding predictions to detect a wider range of cardiovascular conditions, and enhancing noise-handling techniques are all suggested ways to increase the model's clinical applicability and efficacy. The framework needs A robust pre-processing pipeline is necessary since the prediction accuracy depends on high-quality ECG signal images, and missing or erroneous data can significantly lower that accuracy. Furthermore, real-time or resource-constrained scenarios could lack the computational capacity required to manage the computational complexity of hybrid deep learning models, especially when combining various architectures.

Data availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions

Sivabalaselvamani Dhandapani: Conceptualization, Methodology and Implementation, Writing Draft, Review and Editing; Hemalatha Somasundaram: Conceptualization, Formal analysis, Methodology and Implementation, Writing—original draft, Writing—review and editing; Tamilarasi Angamuthu: Supervision, Writing Draft, Review and Editing; All the authors read and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to S.D.

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